

MINI-PROJECT 2 SHORT TECHNICAL REPORT

Course: Cross-Platform Mobile App Development (VKU)
Title: Real-time Study Room Booking App (React Native & Expo)
Weeks: 5-6 Weight: 10%
Student: Nguyễn Minh Duy — 23IT038 — abc@vku.udn.vn
Date: 01/10/2026

1. GENERAL INFORMATION & DELIVERABLE LINKS

Solo project. Nguyễn Minh Duy (23IT038) built the Expo app, booking rules, Cloudflare KV API, and this report. Contribution: 100%.

Live Demo URL: <https://vku-room-booking-part2.pages.dev/>
API health: <https://vku-room-booking-part2.pages.dev/api/health>
GitHub: <https://github.com/minhduy6868/mob-lab-room-booking>
Video: Not recorded. The live URL is the demo deliverable.

Demo login is pre-filled: abc@vku.udn.vn / vku@2026 (Nguyễn Minh Duy, 23IT038). Week 5 remains at <https://vku-room-booking.pages.dev/>. Week 6 is the part2 host above.

2. FEATURE IMPLEMENTATION CHECKLIST

#	Required feature	Status	Where it lives
1	Room feed: photo, building/floor, capacity, Available vs Occupied	Complete	FlatList in BrowseRoomsScreen. RoomCard is memoized. windowSize 5, initialNumToRender 10. Status comes from KV bookings for the selected day.
2	Search and chips: building A/B/C/V, capacity 2-20, equipment	Complete	Search matches name, building, amenity, description. Chips: type, building (A, B, C, V, library, dorm), capacity 2-20 or over 20, projector / whiteboard / high-spec PC / AC, available vs full.
3	7-day selector and 2-hour slots with conflicts disabled	Complete	DateStrip shows 7 days. Slots: 07:30-09:30, 09:45-11:45, 13:00-15:00, 15:15-17:15, 17:30-19:30, 19:45-21:45. Occupied, past, overlap, daily limit (2), and maintenance slots are disabled. Rules run in booking-rules.ts and functions/_lib/rules.js.
4	Booking pass with QR check-in	Complete	POST /api/bookings returns a booking code. BookingConfirmation is a modal pass. QRCodePassModal opens from My Bookings. Check-in is allowed from 10 minutes before the slot.
5	Zustand: session, reservations, filters, cancel	Complete	useAuthStore holds the session. useBookingStore holds bookings, date, filters, book, cancel, and check-in. TanStack Query holds the room catalog and the live booking cache.
6	Persist with AsyncStorage	Complete	Zustand persist uses @react-native-async-storage/async-storage (localStorage on web). Auth session and booking/filter cache are restored on the next launch. KV remains the source of truth.
7	Check-in reminder 15 minutes before the slot	Complete on device	expo-notifications schedules a local alert 15 minutes before start and cancels it if the booking is cancelled. The Pages web demo does not show an OS notification; Expo Go / a dev build does.
8	Navigation, motion, list performance	Complete	Root stack + bottom tabs. Typed params for RoomDetails and BookingConfirmation. Reanimated FadeInDown on cards. Swipe-left cancel uses Gesture Handler.

3. TECHNICAL ARCHITECTURE

One Expo SDK 57 TypeScript app. Web export is hosted on Cloudflare Pages. The same screens run in Expo Go. A booking is accepted only after Pages Functions write Cloudflare KV. The client then upserts that record into Zustand and the TanStack Query cache, so the open session updates from the response. Another browser sees the change on focus, reconnect, or pull-to-refresh. There is no 2.5 second poll and no WebSocket.

App.tsx Navigation, Query, alerts; notification handler on iOS/Android
src/navigation/ Root stack + tabs; RoomDetails; BookingConfirmation modal
src/screens/ Login, Browse, RoomDetails, MyBookings, Profile
src/store/useAuthStore.ts Session
src/store/useBookingStore.ts Filters, book, cancel, check-in, reminder hook
src/lib/booking-rules.ts Conflict engine (client)
src/lib/reminders.ts 15-minute local notification
src/lib/storage.ts AsyncStorage adapter
functions/api/bookings.js GET, POST, PUT cancel|checkin
functions/_lib/rules.js Same conflict rules on the server
wrangler.toml KV binding VKU_BOOKINGS

Flow: login -> hydrate bookings from GET /api/bookings -> pick a day and a free slot -> POST /api/bookings. The server returns 200 plus a booking code, or 409 plus a Vietnamese conflict message. Cancel is PUT with action cancel, and is refused inside 30 minutes of the start time.

4. SCREENSHOTS

Captured from <https://vku-room-booking-part2.pages.dev/> at 390x844. Login used the pre-filled demo account.

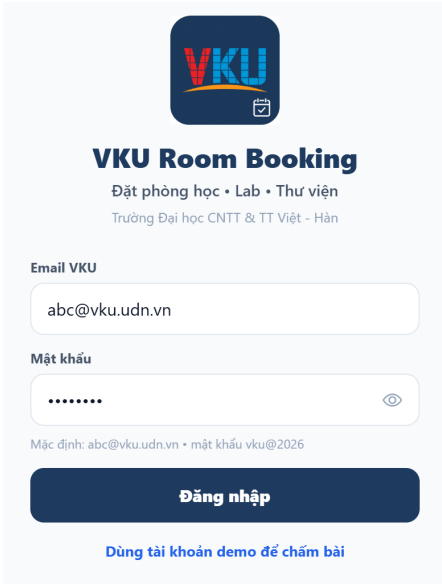


Figure 1. Login. Email `abc@vku.udn.vn` and password are pre-filled.

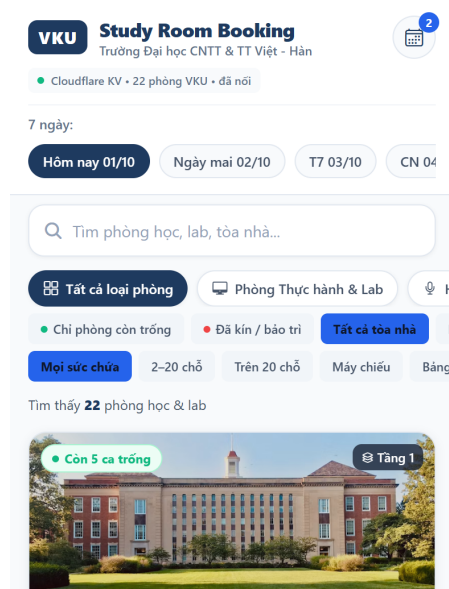


Figure 2. Browse. 7-day strip, building/capacity/equipment cards.



Figure 3. Room details. Slot list disables occupied and past slots.



Figure 4. My Bookings. Pull to refresh. Swipe left cancels a valid booking.

5. WORK BEYOND THE BRIEF

The brief requires a fast Expo list, Zustand, slot conflicts, and a 15-minute reminder. This build adds the following, each tied to a file a reader can open.

Typed navigation. RootNavigator is a native stack over bottom tabs. RoomDetails receives { roomId, roomName }. BookingConfirmation receives { bookingId } and is presented as a modal, so the pass covers the tab bar. Route params are RootStackParamList, not untyped strings.

Server state is separate from UI state. useAuthStore and useBookingStore keep the session, filters, and the open booking. TanStack Query keeps the room catalog and the booking cache. After POST or PUT, the JSON booking returned by Cloudflare is upserted into both Zustand and the query cache. The screen updates from that response. A second browser catches up on focus, reconnect, or pull-to-refresh. The old 2.5 second poll is gone.

The conflict engine runs twice. `src/lib/booking-rules.ts` blocks a taken slot, a personal overlap, a third slot on the same day, a past slot, and a room in maintenance before the request is sent. `functions/_lib/rules.js` applies the same rules on POST, so another client cannot skip the UI. Cancel is refused inside 30 minutes of the start. Check-in opens 10 minutes before the slot.

Motion stays on the UI thread. Room cards use `Reanimated FadeInDown`, `FadeOutUp`, and a spring layout. Booking cards use `Gesture.Pan`: a left swipe past 120px opens the same cancel confirmation as the button. `FlatList` windowing (`windowSize 5`, `initialNumToRender 10`) and a memoized `RoomCard` keep the 22-room feed from re-rendering every card on each keystroke.

6. LIMITS

Cloudflare KV `list()` can lag a few seconds after a write, so a second device may not see a new booking until focus or pull-to-refresh. `expo-notifications` schedules the 15-minute check-in alert on iOS and Android; the Pages web demo does not show an operating-system notification. Public repository: <https://github.com/minhduy6868/mob-lab-room-booking>